GradientCIO

PORTFOLIO ATTRIBUTION COMPARISON

Northstar Total Fund Attribution Comparison

Historical and forward attribution · available evidence review

PORTFOLIO
Northstar Total Fund

BENCHMARK
Policy benchmark

COHORT
Total fund

AVAILABLE VIEWS
Historical and forward

01 Executive Comparison

Two governed lanes are available without server comparison metrics

Two attribution views available

BOTTOM LINE

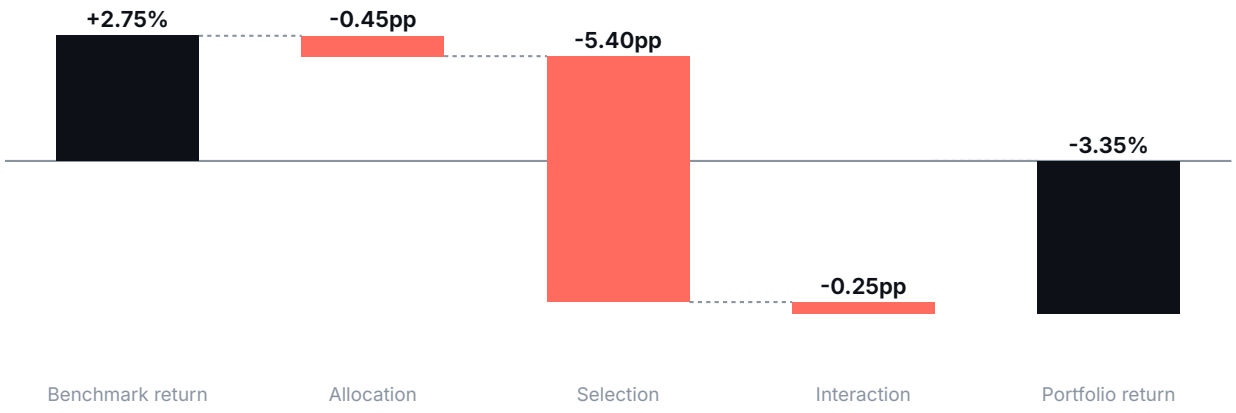
Realized shortfall comes from selection; expected gain comes from selection; Alternatives drives both S2 S3.

HISTORICAL LANE	EX ANTE LANE	SIMULATED LANE	COMPARISON COVERAGE
Available	Available	Unavailable	Fallback
Realized 12 months · 2025-10-31 to 2026-09-30 S2	1-year expected S3	Not available — planned per get_gradient_capabilities S1	Not available — planned per get_gradient_capabilities S1

Realized shortfall from selection; expected gain from selection; Alternatives drives both S2 S3.

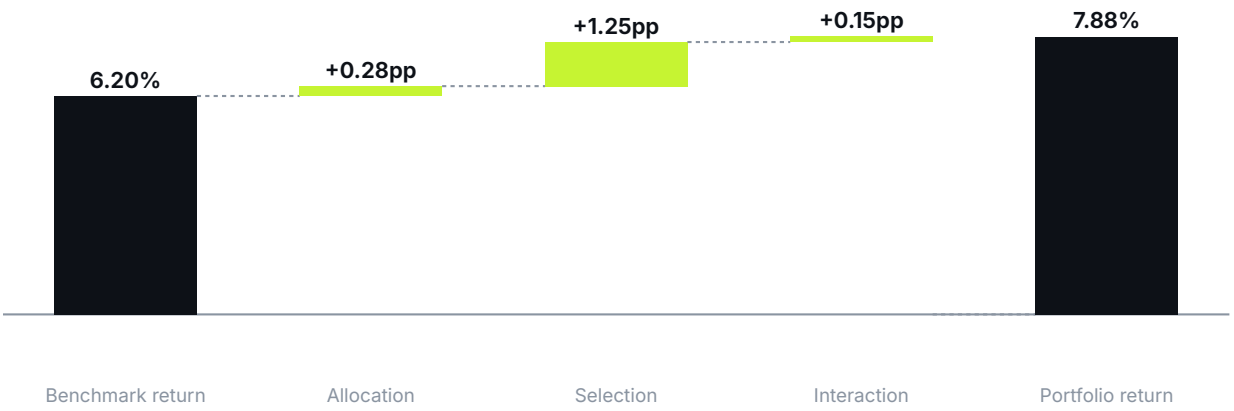
Returned evidence only; no local subtraction or accumulation.

REALIZED SUMMARY EFFECTS



Each plotted value is a returned summary field S2.

EXPECTED SUMMARY EFFECTS



Each plotted value is a returned summary field S3.

Key judgment — Historical

Observation: Selection is the dominant returned realized shortfall, driven by Alternatives S2. Why it matters: It bridges the +2.75% benchmark return to the -3.35% portfolio return. Uncertainty: One realized window may not persist. What would change the view: A later governed period with a different returned selection effect.

Key judgment — Ex ante

Observation: Selection is the dominant returned expected gain, driven by Alternatives S3. Why it matters: It bridges the 6.20% expected benchmark return to the 7.88% expected portfolio return. Uncertainty: Inputs are governed assumptions. What would change the view: Refreshed inputs with a different returned selection effect.

Key judgment — Comparison

Observation: Comparison capability is unavailable S1. Why it matters: Persistence cannot be assessed. Uncertainty: A future tool may return metrics. What would change the view: Published capability.

02 Scope and Compatibility

Portfolio, benchmark, cohort and currency identifiers match

Compatible for display-only fallback

Portfolio, benchmark role, parent cohort and calculation currency match S2 S3.

Compatibility gate	● **Available**	Four identifiers match S2 S3

03 Historical Attribution

Returned realized effects are shown without recalculation

HISTORICAL EFFECTS BY SEGMENT



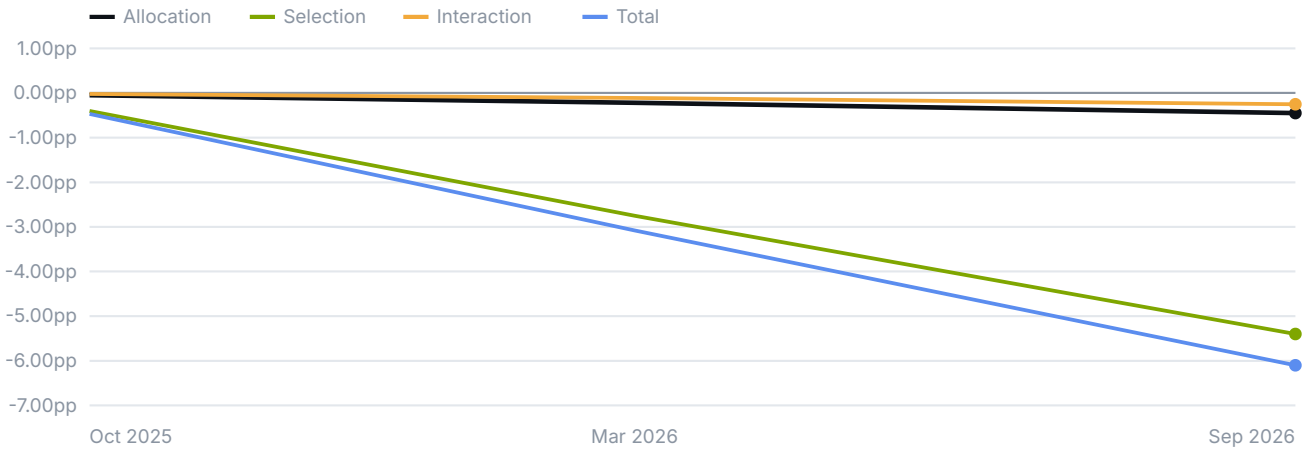
Returned allocation, selection and interaction effects by segment S2.

HISTORICAL PORTFOLIO AND BENCHMARK AVERAGE WEIGHTS



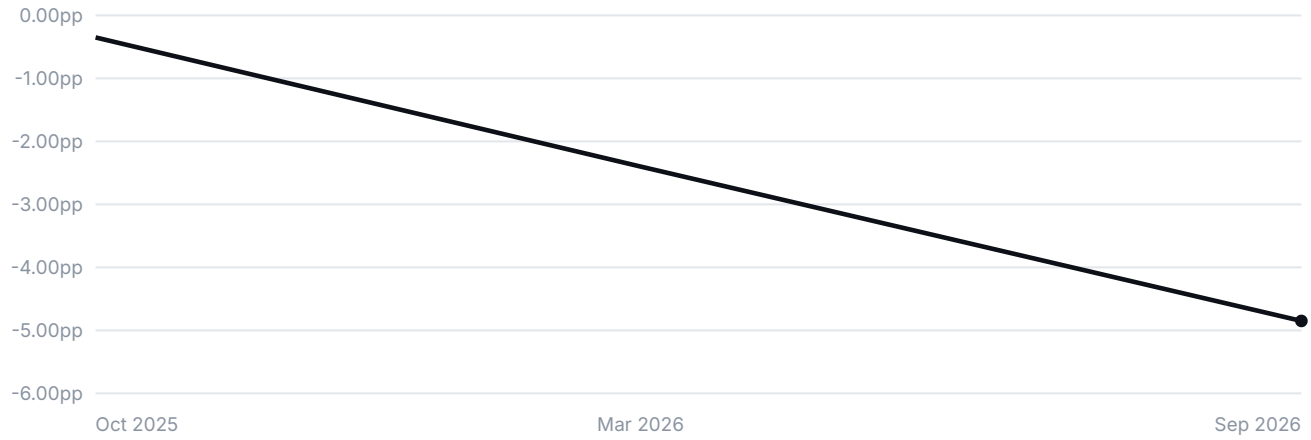
Returned arithmetic average beginning weights; no local difference S2.

CUMULATIVE LINKED EFFECTS



Server-returned cumulative linked fields; no local accumulation S2.

CUMULATIVE ALTERNATIVES SELECTION



Returned rows filtered to allocation_name=Alternatives; no local accumulation S2.

Returned cumulative growth context

Not available — cumulative_growth was not returned.

Analysis — Selection leads detractors

Observation: Returned selection is the -5.40pp dominant realized shortfall and Alternatives contributes -4.85pp S2. Why it matters: Selection drives the returned bridge from +2.75% benchmark return to -3.35% portfolio return. Uncertainty: The evidence covers one linked window. What would change the view: A later historical result with a different returned selection effect.

Benchmark and excess boundary

Returned diagnostics.entity_return_reconciliation keeps attribution-summary excess against the linked segment benchmark composite separate from historical-return excess against the policy benchmark entity S2.

04 Governed Ex Ante Attribution

Returned expected effects are shown as assumptions

EXPECTED EFFECTS BY SEGMENT



Returned expected allocation, selection and interaction by segment S3.

EX ANTE ASSUMPTIONS

assumption_set_name	**2026 Strategic CMA** S3
regime_name	**user-defined regime** S3
horizon	**1 year** S3
portfolio_return_source	**allocation_expected_return_base** S3
benchmark_return_source	**benchmark_expected_return_base** S3

EXPECTED PORTFOLIO AND BENCHMARK RETURNS



Returned expected returns grouped without local subtraction S3.

> **Analysis — Selection leads expectations**
> Observation: Returned selection is the +1.25pp dominant expected gain and Alternatives contributes +1.05pp S3. Why it matters: Selection drives the returned bridge from 6.20% expected benchmark return to 7.88% expected portfolio return. Uncertainty: Inputs are governed assumptions. What would change the view: A refreshed ex ante result with a different returned selection effect.

05 Simulated Attribution

Simulation capability is planned

> **Not available.** Planned per get_gradient_capabilities S1.

Capability	**Not available**	Planned per capability source S1
Invocation	**Not available**	Not called because capability is unavailable S1

> **Analysis — Simulation unavailable**
> Observation: Simulation is planned per the capability source S1. Why it matters: Scenario effects cannot be assessed. Uncertainty: Capability readiness may change. What would change the view: A capability response marking the exact simulation tool available.

06 Cross-Lane Comparison

Returned lanes are displayed while comparison remains planned

> **Not available.** Planned per get_gradient_capabilities S1.

RETURNED CROSS-LANE EFFECT MATRIX

	REALIZED	EXPECTED
ALLOCATION	0.20	0.40
SELECTION	-0.80	-0.10
INTERACTION	0.00	0.05

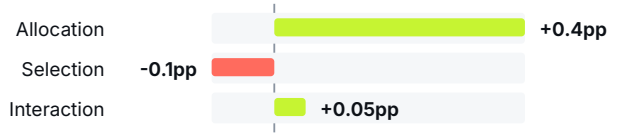
Side-by-side returned summary fields; no local differences S2 S3.

REALIZED RETURNED EFFECTS



Displayed unchanged S2.

EXPECTED RETURNED EFFECTS



Displayed unchanged S3.

Unavailable comparison metrics

Differences, concentration, persistence, and Sources that persist are unavailable without a governed comparison result S1.

Lane differences	**Not available**	Requires governed comparison output S1
Concentration	**Not available**	No local aggregation permitted S1
Persistence	**Not available**	No server persistence metric S1
Sources that persist	**Not available**	No governed comparison result S1

Analysis — Comparison unavailable

Observation: Comparison is planned per get_gradient_capabilities S1. Why it matters: No persistence claim is supported. Uncertainty: Planned capability timing is not returned. What would change the view: A governed comparison result.

07 Coverage and Limitations

Historical and ex ante lanes returned; simulation and comparison are not yet available

Historical	**Available**	Realized attribution returned S2
Ex ante	**Available**	Expected attribution returned S3
Simulation and comparison	**Not available**	Not yet available on this connector S1

A Appendix — Sources and methods

Governed source inventory

Based on 2 of 4 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Capability gate	get_gradient_capabilities	2026-09-30	Passed
S2	Historical attribution	get_portfolio_attribution	2026-09-30	Passed
S3	Ex ante attribution	get_portfolio_ex_ante_attribution	2026-09-30	Passed

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	DIGEST
S1	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S2	sha256:bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb
S3	sha256:ccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

RETURNED SIGN BY SEGMENT

LANE	SEGMENT	RETURNED EFFECT	RETURNED SIGN	SOURCE
Historical	Public Equity	+0.2pp	Positive	S2
Historical	Alternatives	-0.8pp	Negative	S2
Governed ex ante	Public Equity	+0.27pp	Positive	S3
Governed ex ante	Alternatives	+0.08pp	Positive	S3

Signs classify returned effects only; no cross-lane arithmetic S2 S3.

Method boundary

Each lane keeps its own method, period, benchmark role and currency, and values are shown as returned; a missing simulation or comparison capability is a limitation, not a zero value S1 S2 S3.